

Qualification Pack



Rare Earths Extraction Plant Operator

QP Code: MIN/Q1806

Version: 1.0

NSQF Level: 4.0

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MIN/Q1806: Rare Earths Extraction Plant Operator

Brief Job Description

A Rare Earths Extraction Plant Operator is responsible for conducting the extraction of rare earth using appropriate equipment and following the applicable processes including health and safety at the worksite.

Personal Attributes

This job requires the individual to concentrate on the task at hand and complete it efficiently and effectively. The individual should also be physically agile, strong, have good eye sight and not suffer from colour-blindness.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [MIN/N1817: Use different material handling equipment in rare earth extraction plant](#)
2. [MIN/N1818: Undertake different unit operations in rare earth extraction plant](#)
3. [MIN/N1819: Undertake the treatment of rare earth extraction plant effluents](#)
4. [MIN/N1820: Follow the Material Safety Data Sheet \(MSDS\) for chemicals](#)
5. [MIN/N1821: Carry out corrective and preventive maintenance of RE extraction plant equipment](#)
6. [MIN/N1810: Follow the applicable health, safety and environmental guidelines \(Placer Mining\)](#)
7. [MIN/N1811: Work collaboratively and effectively in a team](#)
8. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Mining
Sub-Sector	Mining Operation
Occupation	Placer Mining
Country	India
NSQF Level	4.0

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Credits	14
Aligned to NCO/ISCO/ISIC Code	NCO-2015/ 3117.9900,
Minimum Educational Qualification & Experience	12th grade Pass OR Completed 2nd year of the 3-year diploma after 10 (in Mining/ Chemical/ Metallurgy/ Material Science) OR 10th grade pass (with two years of NTC/ NAC/ CITS or equivalent (in Mechanical/ Electrical)) OR 11th grade pass (with 1 year relevant experience) OR 10th grade pass (with 2 years relevant experience) OR 8th grade pass with 2 year NTC plus 1 year NAC plus 1 year CITS (in Mechanical) OR Previous relevant Qualification of NSQF Level (3.0)(with 3 years relevant experience) OR Previous relevant Qualification of NSQF Level (3.5)(with 1.5 year relevant experience)
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	14/03/2027
NSQC Approval Date	15/03/2024
Version	1.0
Reference code on NQR	QG-04-MG-02174-2024-V1-SCMS
NQR Version	1.0

Remarks:

420 Hours

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MIN/N1817: Use different material handling equipment in rare earth extraction plant

Description

This unit is about conveying the monazite sand to the processing plant through pipeline systems and conveyors

Scope

The scope covers the following :

- Operation of belt conveyors and bucket elevator
- Operation of screw conveyors
- Operation of pumps and pipelines
- Operation of valves

Elements and Performance Criteria

Operation of belt conveyors and bucket elevator

To be competent, the user/individual on the job must be able to:

- PC1.** check the material handling equipment, such as bucket elevator, bucket belt conveyor, belt conveyor, screw conveyors etc., for the correct functioning
- PC2.** check the layout and components of the belt conveyor system including its principal components, i.e. conveyor belt, clips, roller chains, guards, tensioners, feed and discharge chutes, head, tail and take-up pulleys, bearings, braking system, controls such as switches, trip cords, interlocks, alarms, etc.
- PC3.** check the mechanical and electrical driving units including safety devices for effective functioning
- PC4.** convey the monazite sand to the processing unit considering appropriate parameters, such as feed/ material rate, alignment of conveyor belts, spillage amount and rate, conveyor jams, etc.
- PC5.** inspect the equipment for potential issues, e.g. unusual noises, blockages and obstruction, and the requirement for lubricating motors, conveyor and electromagnet
- PC6.** check the hoppers and bins, chutes, and guardrails deployed to control stockpile levels

Operation of screw conveyors

To be competent, the user/individual on the job must be able to:

- PC7.** check the layout and components of screw conveyors including conveyor screw, trough, hanger bearings, couplings, internal collar, end lugs, shafts, feed/discharge chute, drive mechanisms, lubrication systems, controls, drive belts, hangers/cables, etc., for wear and tear or damage
- PC8.** feed material to the various processing equipment, maintaining the recommended parameters, e.g. feed rate, voltage, Revolution per Minute (RPM), etc.
- PC9.** identify safety hazards, such as spillage, leaks, abnormal noises and odour, and take appropriate preventive and corrective measures

Operation of pumps and pipelines

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To be competent, the user/individual on the job must be able to:

- PC10.** transfer slurry and other chemicals through pipelines
- PC11.** check the water pumps, oil pumps and vacuum pumps for the correct functioning, and carry out appropriate troubleshooting
- PC12.** check the PCD of pipelines, flanges and their fitting, and different types of fasteners used for pipelines
- PC13.** identify faults in the machinery and make suitable adjustments to the machinery to improve their performance
- PC14.** identify common safety hazards concerning the use of pumps and pipelines and take appropriate preventive and corrective measures, as applicable

Operation of valves

To be competent, the user/individual on the job must be able to:

- PC15.** check different types of valves and their components for wear and tear or damage, and take appropriate corrective measures
- PC16.** control the feeding of chemicals to the processing equipment using valves

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** different types of unit operations in a rare earth extraction plant
- KU2.** the use of different types of pumps and conveyors, such as belt types, screw types, pneumatic types etc.
- KU3.** the maintenance of pipeline systems, pumps, motors, conveyors and the principal conveyor components, such as screws, screw casing/trough scrapers, conveyor belt, splices or clips, drive mechanisms, roller chains/belts, buckets, guards, tensioners, feed and discharge chutes, head, tail and take-up pulleys, idlers, bearings, braking system, controls (such as switches, trip cords, side travel switches, interlocks, crossovers, audible alarms, magnet, metal detectors, and lubricating systems
- KU4.** different types of screw conveyors, e.g. horizontal screw conveyors, inclined screw conveyors, vertical screw conveyors, shaft-less screw conveyors, etc.
- KU5.** the principal components of screw conveyors, e.g. conveyor screw, trough, hanger bearings, couplings, internal collar, end lugs, shafts, feed/discharge chute, drive mechanisms, lubrication systems, controls, drive belts, hangers/cables, etc.
- KU6.** the duties of workmen at a rare earth extraction plant
- KU7.** the provision of wages, working hours and accident compensation as per the Mines act
- KU8.** the specific safety instructions for conveying systems (pull chord, label switch, fire alarm etc.)
- KU9.** the belt mechanisms/systems (belt fastening, belt jointing etc.)
- KU10.** the hazard and safety aspects concerning ore processing activities, and the usage of relevant Personal Protective Equipment (PPE)

Generic Skills (GS)

User/individual on the job needs to know how to:



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- GS1.** interpret the sketches and engineering drawings and note down observations, if any
- GS2.** prepare draft drawings for the final output product
- GS3.** enter the information in online ERP systems under the guidance of the supervisor
- GS4.** read the equipment manuals and process documents for a better understanding of equipment and processes
- GS5.** discuss task lists, schedules and activities with the supervisor
- GS6.** communicate effectively with the team members
- GS7.** plan and organize work to ensure timely completion
- GS8.** organize the process/ equipment manuals/ documents for convenient sorting/ accessing of information
- GS9.** apply common sense to make appropriate judgments concerning work
- GS10.** follow instructions and work on the identified areas of improvement
- GS11.** complete the assigned tasks within the applicable time limits with minimum supervision
- GS12.** take appropriate initiatives to drive small projects with the supervisor, e.g. training and operation improvement of helpers, assistant operators, etc.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Operation of belt conveyors and bucket elevator</i>	10	17	-	8
PC1. check the material handling equipment, such as bucket elevator, bucket belt conveyor, belt conveyor, screw conveyors etc., for the correct functioning	2	3	-	2
PC2. check the layout and components of the belt conveyor system including its principal components, i.e. conveyor belt, clips, roller chains, guards, tensioners, feed and discharge chutes, head, tail and take-up pulleys, bearings, braking system, controls such as switches, trip cords, interlocks, alarms, etc.	1	3	-	1
PC3. check the mechanical and electrical driving units including safety devices for effective functioning	1	3	-	1
PC4. convey the monazite sand to the processing unit considering appropriate parameters, such as feed/ material rate, alignment of conveyor belts, spillage amount and rate, conveyor jams, etc.	2	2	-	1
PC5. inspect the equipment for potential issues, e.g. unusual noises, blockages and obstruction, and the requirement for lubricating motors, conveyor and electromagnet	2	3	-	1
PC6. check the hoppers and bins, chutes, and guardrails deployed to control stockpile levels	2	3	-	2
<i>Operation of screw conveyors</i>	6	10	-	3
PC7. check the layout and components of screw conveyors including conveyor screw, trough, hanger bearings, couplings, internal collar, end lugs, shafts, feed/discharge chute, drive mechanisms, lubrication systems, controls, drive belts, hangers/cables, etc., for wear and tear or damage	2	4	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC8. feed material to the various processing equipment, maintaining the recommended parameters, e.g. feed rate, voltage, Revolution per Minute (RPM), etc.	2	3	-	1
PC9. identify safety hazards, such as spillage, leaks, abnormal noises and odour, and take appropriate preventive and corrective measures	2	3	-	1
<i>Operation of pumps and pipelines</i>	10	17	-	7
PC10. transfer slurry and other chemicals through pipelines	2	3	-	1
PC11. check the water pumps, oil pumps and vacuum pumps for the correct functioning, and carry out appropriate troubleshooting	2	3	-	1
PC12. check the PCD of pipelines, flanges and their fitting, and different types of fasteners used for pipelines	2	3	-	1
PC13. identify faults in the machinery and make suitable adjustments to the machinery to improve their performance	2	4	-	2
PC14. identify common safety hazards concerning the use of pumps and pipelines and take appropriate preventive and corrective measures, as applicable	2	4	-	2
<i>Operation of valves</i>	4	6	-	2
PC15. check different types of valves and their components for wear and tear or damage, and take appropriate corrective measures	2	3	-	1
PC16. control the feeding of chemicals to the processing equipment using valves	2	3	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1817
NOS Name	Use different material handling equipment in rare earth extraction plant
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Placer Mining
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

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MIN/N1818: Undertake different unit operations in rare earth extraction plant

Description

This unit is about undertaking different rare earth extraction plant unit operations, e.g. grinding, attacking, leaching, crystallizing, centrifuging, etc

Scope

The scope covers the following :

- Grinding operation
- Attack operation
- Leaching process
- Utilities
- Crystallization process
- Centrifuging operation
- TSP flash drying operation
- Evaporation process
- RE extraction process
- Deactivation process
- RE evaporation process
- Filtration

Elements and Performance Criteria

Grinding operation

To be competent, the user/individual on the job must be able to:

- PC1.** follow the working principle of the ball mill, considering parameters like feed/ material rate, residence time
- PC2.** follow the standard procedure for the startup and shut down of the ball mill unit
- PC3.** using rope/barriers, signboard, etc., to minimize the relevant hazards and report the condition to the supervisor
- PC4.** identify spillage, blockages, obstruction etc., and take appropriate corrective measures

Attack operation

To be competent, the user/individual on the job must be able to:

- PC5.** follow the standard procedure of agitator unit startup and shutdown
- PC6.** maintain the relevant process parameters during the addition of caustic lye and steam
- PC7.** ensure the availability of guards and fences against potential hazards like spills, leaks and overflows

Leaching process

To be competent, the user/individual on the job must be able to:

- PC8.** follow the leaching process appropriately
- PC9.** monitor the working of the pedal type of agitator, and decantation operation

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PC10. maintain the relevant process parameters during the addition of wash water and other chemicals

PC11. control the temperature, agitation time, and settling time, as required

Utilities

To be competent, the user/individual on the job must be able to:

PC12. follow the appropriate cooling tower, boiler plant, DM Plant, and water handling processes

PC13. monitor the compressor and air dryer operation and maintain the relative process parameters, such as inlet pressure, temperature, etc

Crystallization process

To be competent, the user/individual on the job must be able to:

PC14. monitor the process of crystallization through adiabatic cooling

PC15. maintain the relevant process parameters during crystallization

Centrifuging operation

To be competent, the user/individual on the job must be able to:

PC16. monitor the process of centrifuging and maintain the relevant process parameters during centrifuging

PC17. follow the standard procedure of centrifuging unit startup and shutdown

TSP flash drying operation

To be competent, the user/individual on the job must be able to:

PC18. monitor the process of TSP centrifuging and drying, maintaining the relevant process parameters

PC19. follow the standard procedure of dryer unit startup and shutdown

PC20. maintain a minimum of eight air changes per TSP bagging area

PC21. conduct operational checks for potential issues, e.g. noises, blockages, obstruction, etc., during drying and take appropriate corrective measures

Evaporation process

To be competent, the user/individual on the job must be able to:

PC22. follow the standard procedure of the evaporator startup and shutdown

PC23. monitor the evaporation process through vacuum and maintain the process parameters

RE extraction process

To be competent, the user/individual on the job must be able to:

PC24. monitor the operation of pitch blade type agitator, filter press

PC25. monitor the working of the diaphragm pump, ODS pump, centrifugal pump and metering pump

PC26. maintain the relevant process parameters during the addition of different types of chemicals

PC27. follow the standard procedure of filter press unit startup and shutdown

Deactivation process

To be competent, the user/individual on the job must be able to:

PC28. monitor the operation of the turbo disc agitator

PC29. monitor the working of the diaphragm pump

PC30. maintain the relevant process parameters during the addition of the required quantity of chemicals and residue

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RE evaporation process

To be competent, the user/individual on the job must be able to:

- PC31.** monitor the operation of the vat, pump, agitator, and Flaker unit
- PC32.** monitor the procedure for charging steam and handling condensate
- PC33.** ensure proper packing and storage of the finished product

Filtration

To be competent, the user/individual on the job must be able to:

- PC34.** conduct filtration of RE mixture by controlling the moisture, filtrate, pressure, feed rate, etc.
- PC35.** check the condition of the filtration systems including filter cloth, hydraulic systems
- PC36.** monitor the feed pressure, air pressure and moisture in the exhaust air
- PC37.** conduct operational checks for potential issues, e.g. unusual leaks, noises, blockages and obstruction, etc., during filtration and take appropriate corrective measures
- PC38.** perform air drying of filtered cake at an appropriate pressure for the recommended duration

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** different types of chemical plants and their operations
- KU2.** the importance of plant organization, timekeeping, discipline and punctuality
- KU3.** the requirement of fencing, guarding, and spillage control measures
- KU4.** the importance of first aid and hygiene
- KU5.** the standing orders in force at the factory/plant
- KU6.** the provision for wages, working hours and accident compensation as per Atomic Energy Factory Rule (AEFR)
- KU7.** the impact of violation of safety procedures
- KU8.** the relevant organizational standards and procedures concerning procurement, store management, inventory management, quality management, etc.
- KU9.** the key contact points for query resolution
- KU10.** different types of unit operations in rare earth extraction processes and associated equipment
- KU11.** different process parameters; types of reagents like acids, alkalis, solvents, flocculants etc., and their mixing proportions
- KU12.** different types of alarms, pumps, filters, etc.
- KU13.** different types of filtration systems and their components, such as plates, frames, hydraulic systems, filter cloth, air drying systems, slurry recirculation system
- KU14.** the functioning of equipment, such as metering pumps, centrifugal pumps, diaphragm pumps, etc.
- KU15.** the checking and testing procedure for rare earth extraction equipment

Generic Skills (GS)

User/individual on the job needs to know how to:



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- GS1.** read the parameters on various types of monitoring panels
- GS2.** note measurements, equipment panel readings for various process parameters in the required reporting formats
- GS3.** discuss task lists, schedules and activities with the supervisor
- GS4.** communicate effectively with the team members and instruct the subordinates to achieve the completion of allocated tasks
- GS5.** listen attentively with full attention and comprehend the information given by the speaker
- GS6.** plan and organize work to ensure timely completion
- GS7.** support the supervisor in scheduling tasks for subordinates, e.g. helper and assistant operator
- GS8.** apply common sense to make appropriate judgments concerning work
- GS9.** use intuition to detect potential work-related problems and take appropriate preventive or corrective measures through discussion with the supervisor

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Grinding operation</i>	4	6	-	3
PC1. follow the working principle of the ball mill, considering parameters like feed/ material rate, residence time	1	2	-	1
PC2. follow the standard procedure for the startup and shut down of the ball mill unit	1	1	-	1
PC3. using rope/barriers, signboard, etc., to minimize the relevant hazards and report the condition to the supervisor	1	1	-	-
PC4. identify spillage, blockages, obstruction etc., and take appropriate corrective measures	1	2	-	1
<i>Attack operation</i>	3	4	-	1
PC5. follow the standard procedure of agitator unit startup and shutdown	1	2	-	1
PC6. maintain the relevant process parameters during the addition of caustic lye and steam	1	1	-	-
PC7. ensure the availability of guards and fences against potential hazards like spills, leaks and overflows	1	1	-	-
<i>Leaching process</i>	3	6	-	2
PC8. follow the leaching process appropriately	1	1	-	-
PC9. monitor the working of the pedal type of agitator, and decantation operation	1	2	-	1
PC10. maintain the relevant process parameters during the addition of wash water and other chemicals	1	2	-	1
PC11. control the temperature, agitation time, and settling time, as required	-	1	-	-
<i>Utilities</i>	2	2	-	-
PC12. follow the appropriate cooling tower, boiler plant, DM Plant, and water handling processes	1	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. monitor the compressor and air dryer operation and maintain the relative process parameters, such as inlet pressure, temperature, etc	1	1	-	-
<i>Crystallization process</i>	1	2	-	1
PC14. monitor the process of crystallization through adiabatic cooling	1	1	-	-
PC15. maintain the relevant process parameters during crystallization	-	1	-	1
<i>Centrifuging operation</i>	1	2	-	-
PC16. monitor the process of centrifuging and maintain the relevant process parameters during centrifuging	1	1	-	-
PC17. follow the standard procedure of centrifuging unit startup and shutdown	-	1	-	-
<i>TSP flash drying operation</i>	4	5	-	2
PC18. monitor the process of TSP centrifuging and drying, maintaining the relevant process parameters	1	1	-	1
PC19. follow the standard procedure of dryer unit startup and shutdown	1	1	-	1
PC20. maintain a minimum of eight air changes per TSP bagging area	1	2	-	-
PC21. conduct operational checks for potential issues, e.g. noises, blockages, obstruction, etc., during drying and take appropriate corrective measures	1	1	-	-
<i>Evaporation process</i>	1	2	-	1
PC22. follow the standard procedure of the evaporator startup and shutdown	1	1	-	-
PC23. monitor the evaporation process through vacuum and maintain the process parameters	-	1	-	1
<i>RE extraction process</i>	3	5	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. monitor the operation of pitch blade type agitator, filter press	-	1	-	1
PC25. monitor the working of the diaphragm pump, ODS pump, centrifugal pump and metering pump	1	1	-	1
PC26. maintain the relevant process parameters during the addition of different types of chemicals	1	2	-	-
PC27. follow the standard procedure of filter press unit startup and shutdown	1	1	-	-
<i>Deactivation process</i>	3	5	-	2
PC28. monitor the operation of the turbo disc agitator	1	2	-	1
PC29. monitor the working of the diaphragm pump	1	1	-	-
PC30. maintain the relevant process parameters during the addition of the required quantity of chemicals and residue	1	2	-	1
<i>RE evaporation process</i>	2	3	-	2
PC31. monitor the operation of the vat, pump, agitator, and Flaker unit	1	1	-	1
PC32. monitor the procedure for charging steam and handling condensate	1	1	-	-
PC33. ensure proper packing and storage of the finished product	-	1	-	1
<i>Filtration</i>	3	8	-	4
PC34. conduct filtration of RE mixture by controlling the moisture, filtrate, pressure, feed rate, etc.	-	2	-	1
PC35. check the condition of the filtration systems including filter cloth, hydraulic systems	1	2	-	1
PC36. monitor the feed pressure, air pressure and moisture in the exhaust air	-	1	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC37. conduct operational checks for potential issues, e.g. unusual leaks, noises, blockages and obstruction, etc., during filtration and take appropriate corrective measures	1	2	-	-
PC38. perform air drying of filtered cake at an appropriate pressure for the recommended duration	1	1	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1818
NOS Name	Undertake different unit operations in rare earth extraction plant
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Placer Mining
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

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MIN/N1819: Undertake the treatment of rare earth extraction plant effluents

Description

This unit is about treating and disposing of rare earth extraction plant effluents.

Scope

The scope covers the following :

- Treat and dispose of the RE extraction plant effluents

Elements and Performance Criteria

Treat and dispose of the RE extraction plant effluents

To be competent, the user/individual on the job must be able to:

- PC1.** ensure the effluent from the RE processing plant is sent to the effluent treatment area
- PC2.** treat the effluent using the required chemicals
- PC3.** check the pH level of effluent to ensure it is fit for discharge within the limits set by the Pollution Control Board (PCB)
- PC4.** decant the clear solution to another tank, after the settling process
- PC5.** carry out the filtration of the slurry and discharge the solid waste, as per the discharge norms
- PC6.** check the sample of water being discharged and monitor the level of water in ponds

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** different types of chemical plants and their operations
- KU2.** the importance of time-keeping, discipline and punctuality
- KU3.** various chemical processes carried out in a rare earth extraction plant
- KU4.** the in-force standing orders relevant to effluent management at the company/plant
- KU5.** the impact of violation of applicable safety procedures
- KU6.** the relevant quality norms
- KU7.** different types of unit operations involved in rare earth extraction, chemical processes, and associated equipment
- KU8.** different types of effluent treatment processes and the use of associated equipment
- KU9.** the use of different water purifying and waste treatment agents
- KU10.** the hazards and the appropriate safety measures in acid/ alkali handling activities and usage of relevant PPE

Generic Skills (GS)

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User/individual on the job needs to know how to:

- GS1.** read the parameters on various types of monitoring panels, and internal instructions and equipment specification
- GS2.** document information from the sketches and engineering drawings, and parameters from different types of monitoring panels
- GS3.** assist the supervisor in scheduling tasks for subordinates, e.g. helper and assistant operator
- GS4.** analyze the data available about the site
- GS5.** complete the assigned tasks within the applicable time limits with minimum supervision and maintain quality
- GS6.** identify potential problems and appropriate solutions at work through discussion with the supervisor
- GS7.** make appropriate decisions in case of emergencies, following the Authority matrix defined by the organization

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Treat and dispose of the RE extraction plant effluents</i>	30	50	-	20
PC1. ensure the effluent from the RE processing plant is sent to the effluent treatment area	5	5	-	4
PC2. treat the effluent using the required chemicals	5	5	-	4
PC3. check the pH level of effluent to ensure it is fit for discharge within the limits set by the Pollution Control Board (PCB)	5	10	-	4
PC4. decant the clear solution to another tank, after the settling process	5	10	-	2
PC5. carry out the filtration of the slurry and discharge the solid waste, as per the discharge norms	5	10	-	2
PC6. check the sample of water being discharged and monitor the level of water in ponds	5	10	-	4
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1819
NOS Name	Undertake the treatment of rare earth extraction plant effluents
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Placer Mining
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

MIN/N1820: Follow the Material Safety Data Sheet (MSDS) for chemicals

Description

This unit is about following the material safety data sheet of chemicals.

Scope

The scope covers the following :

- Safety measures for handling chemicals

Elements and Performance Criteria

Safety measures for handling chemicals

To be competent, the user/individual on the job must be able to:

- PC1.** maintain Material Safety Data Sheet (MSDS) for chemicals and solvents, such as Caustic lye, Hydrochloric Acid, Nitric Acid, etc.
- PC2.** ensure the availability of relevant data sheet copies to workers and the occupational health committee at the worksite
- PC3.** check the product and company identification details of the chemicals, e.g. product identifier, manufacturer or distributor name, address, phone number, etc.
- PC4.** check the chemical labels to understand the hazards concerning their handling and use
- PC5.** check the composition/information of ingredients and trade secret claims, e.g. the ingredients of chemicals
- PC6.** identify the symptoms of exposure to hazardous chemicals and take the appropriate first-aid measures/ administer treatment
- PC7.** follow the appropriate measures to deal with fire hazards, e.g. use of appropriate firefighting equipment, and the application of appropriate fire extinguishing techniques
- PC8.** follow the appropriate procedure to deal with the accidental release of chemicals, i.e. containment and cleanup using the appropriate tools and protective equipment
- PC9.** follow the handling and storage list precautions to ensure safe handling and storage, including incompatibilities
- PC10.** follow the exposure controls/ personal protection lists for Permissible Exposure Limits (PELs), Threshold Limit Values (TLVs), and any other recommended exposure limit for the chemical
- PC11.** check the physical and chemical properties of each chemical and identify the chemical stability and possibility of hazardous reactions
- PC12.** check the toxicological information, i.e. routes of exposure; related symptoms, acute and chronic effects, numerical measures of toxicity, ecological information, disposal considerations, etc

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

Qualification Pack

- KU1.** the importance of following the applicable safety procedures/norms and the impact of violating them
- KU2.** the applicable quality norms
- KU3.** different types of unit operations involved in RE extraction, chemical processes, and the use of associated equipment
- KU4.** how to maintain Material Safety Data Sheet (MSDS) for chemicals and solvents
- KU5.** the importance of ensuring the availability of relevant data sheet copies to workers and the occupational health committee at the work site
- KU6.** the importance of checking the chemical labels to understand the hazards concerning their handling and use
- KU7.** the symptoms of exposure to hazardous chemicals and the appropriate first-aid measures to be taken
- KU8.** the appropriate fire mitigation measures and the application of appropriate fire extinguishing techniques
- KU9.** the appropriate procedure to deal with the accidental release of chemicals
- KU10.** the importance of following the handling and storage list precautions to ensure the safe handling and storage of chemicals
- KU11.** the exposure controls/ personal protection lists for Permissible Exposure Limits (PELs), Threshold Limit Values (TLVs) for chemicals
- KU12.** the importance of understanding the physical properties of chemicals and the relevant toxicological information

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain discipline and punctuality at work
- GS2.** document information from the sketches and engineering drawings
- GS3.** read the parameters on various types of monitoring panels, and internal instructions and equipment specifications
- GS4.** assist the supervisor in scheduling tasks for subordinates, e.g. helper and assistant operator
- GS5.** analyze the data available about the site
- GS6.** complete the assigned tasks within the applicable time limits with minimum supervision and maintain quality
- GS7.** identify potential problems and appropriate solutions at work through discussion with the supervisor
- GS8.** make appropriate decisions in case of emergencies, following the Authority matrix defined by the organization

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Safety measures for handling chemicals</i>	30	50	-	20
PC1. maintain Material Safety Data Sheet (MSDS) for chemicals and solvents, such as Caustic lye, Hydrochloric Acid, Nitric Acid, etc.	3	5	-	2
PC2. ensure the availability of relevant data sheet copies to workers and the occupational health committee at the worksite	2	3	-	1
PC3. check the product and company identification details of the chemicals, e.g. product identifier, manufacturer or distributor name, address, phone number, etc.	3	5	-	2
PC4. check the chemical labels to understand the hazards concerning their handling and use	3	5	-	2
PC5. check the composition/information of ingredients and trade secret claims, e.g. the ingredients of chemicals	3	5	-	2
PC6. identify the symptoms of exposure to hazardous chemicals and take the appropriate first-aid measures/ administer treatment	3	5	-	2
PC7. follow the appropriate measures to deal with fire hazards, e.g. use of appropriate firefighting equipment, and the application of appropriate fire extinguishing techniques	2	3	-	2
PC8. follow the appropriate procedure to deal with the accidental release of chemicals, i.e. containment and cleanup using the appropriate tools and protective equipment	2	3	-	1
PC9. follow the handling and storage list precautions to ensure safe handling and storage, including incompatibilities	1	3	-	1
PC10. follow the exposure controls/ personal protection lists for Permissible Exposure Limits (PELs), Threshold Limit Values (TLVs), and any other recommended exposure limit for the chemical	2	3	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. check the physical and chemical properties of each chemical and identify the chemical stability and possibility of hazardous reactions	3	5	-	2
PC12. check the toxicological information, i.e. routes of exposure; related symptoms, acute and chronic effects, numerical measures of toxicity, ecological information, disposal considerations, etc	3	5	-	2
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1820
NOS Name	Follow the Material Safety Data Sheet (MSDS) for chemicals
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Placer Mining
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

MIN/N1821: Carry out corrective and preventive maintenance of RE extraction plant equipment

Description

This unit is about corrective and preventive maintenance on different equipment used in a rare earth extraction plant

Scope

The scope covers the following :

- Maintenance of RE extraction plant equipment

Elements and Performance Criteria

Maintenance of RE extraction plant equipment

To be competent, the user/individual on the job must be able to:

- PC1.** prepare mechanical components for the maintenance of equipment, such as grinding, crystallization, evaporators, centrifuge, flash drier setups, filter press, and mixer settlers, associated with the rare earth extraction plant
- PC2.** carry out greasing and lubrication of pivot points
- PC3.** diagnose, service and repair faults in mechanical systems, such as gears, pumps, agitators, valves, mixers, settlers, fans, etc.
- PC4.** ensure appropriate protective enclosures are provided for rotating parts, such as drives, belts, etc.
- PC5.** follow the recommended safety measures during equipment maintenance, i.e. not performing maintenance on a working machine
- PC6.** conduct appropriate tests on the equipment e.g. load testing and vibration testing after repair and maintenance, to ensure everything is working correctly and safely
- PC7.** carry out predictive, preventive and breakdown maintenance for the processing, conveying and other related support equipment, such as fans, pumps, valves pipelines, gearboxes etc.,
- PC8.** follow the standard equipment maintenance procedure recommended by the equipment manufacturer

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the procedure for preparing mechanical components for the maintenance of rare earth extraction plant equipment
- KU2.** the procedure of diagnosing, servicing and repairing faults in the relevant mechanical systems
- KU3.** the importance of providing appropriate protective enclosures for rotating parts
- KU4.** the appropriate safety measures to be followed during equipment maintenance

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- KU5.** the appropriate tests to be conducted on the equipment after repair and maintenance
- KU6.** the process of carrying out predictive, preventive and breakdown maintenance for the processing, conveying and other related support equipment as the recommendation of the equipment manufacturer
- KU7.** the direct/indirect cost of accidents in a rare earth extraction plant
- KU8.** the local emergency response/ disaster management plan
- KU9.** the cost of equipment and loss for the company from in operational equipment
- KU10.** the applicable safety regulations
- KU11.** the use of different equipment, such as metering, pumps, centrifugal pumps, diaphragm pumps etc

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** interpret the relevant drawings, symbols, measurements and Original Equipment Manufacturer (OEM) manuals and make note of appropriate observations
- GS2.** follow the online Enterprise Resource Planning (ERP) systems under the guidance of the supervisor
- GS3.** plan work effectively by discussing task lists, schedules and activities with the supervisor
- GS4.** communicate effectively with the relevant stakeholders
- GS5.** make appropriate decisions about the area of work and work emergencies
- GS6.** practice effective questioning to extract maximum and relevant information about work
- GS7.** undertake work-related tasks in the appropriate sequence for their timely completion based on priority, practising effective time management
- GS8.** follow 5s to do the work easily
- GS9.** identify potential problems and appropriate solutions at work through reasoning and discussion with the supervisor
- GS10.** complete the assigned tasks within the applicable time limits with minimum supervision, maintaining quality

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintenance of RE extraction plant equipment</i>	30	50	-	20
PC1. prepare mechanical components for the maintenance of equipment, such as grinding, crystallization, evaporators, centrifuge, flash drier setups, filter press, and mixer settlers, associated with the rare earth extraction plant	5	6	-	4
PC2. carry out greasing and lubrication of pivot points	5	6	-	2
PC3. diagnose, service and repair faults in mechanical systems, such as gears, pumps, agitators, valves, mixers, settlers, fans, etc.	2	6	-	2
PC4. ensure appropriate protective enclosures are provided for rotating parts, such as drives, belts, etc.	2	6	-	2
PC5. follow the recommended safety measures during equipment maintenance, i.e. not performing maintenance on a working machine	4	6	-	3
PC6. conduct appropriate tests on the equipment e.g. load testing and vibration testing after repair and maintenance, to ensure everything is working correctly and safely	4	7	-	1
PC7. carry out predictive, preventive and breakdown maintenance for the processing, conveying and other related support equipment, such as fans, pumps, valves pipelines, gearboxes etc.,	4	7	-	3
PC8. follow the standard equipment maintenance procedure recommended by the equipment manufacturer	4	6	-	3
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1821
NOS Name	Carry out corrective and preventive maintenance of RE extraction plant equipment
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Placer Mining
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

MIN/N1810: Follow the applicable health, safety and environmental guidelines (Placer Mining)

Description

This unit is about following the applicable health, safety and environmental guidelines.

Scope

The scope covers the following :

- Maintain personal health and safety
- Assist in hazard management
- Work on equipment and machinery
- Check the first aid box, firefighting and safety equipment
- Report incidents

Elements and Performance Criteria

Maintain personal health and safety

To be competent, the user/individual on the job must be able to:

- PC1.** follow the appropriate hand hygiene measures, i.e., washing hands with EDTA/Soda Ash (Ethylene Diamine Tetra Acetate/Sodium Carbonate), soap and water or alcohol-based hand sanitizers
- PC2.** ensure the plants are provided with a dilution ventilation system to avoid the undue accumulation of chemical vapours in working environments
- PC3.** clean and disinfect the regularly used objects, such as mobile phones, wireless communication systems keys, wallets and work passes
- PC4.** follow the Safety Management Plan (SMP) and Emergency Management Plan (EMP)
- PC5.** follow sanitation measures in the work area, e.g., avoiding eating, drinking, chewing and smoking
- PC6.** use the appropriate PPE, e.g. using an appropriate mask to cover the mouth and nose
- PC7.** identify six directional hazards at the workplace and take decisions accordingly
- PC8.** follow appropriate safety measures to avoid exposure to radiation, e.g. limiting stay in the radiation area and avoiding working with open wounds or exposed body parts in radiation areas
- PC9.** undertake the Take-5 (Personal Risk Assessment) before the commencement of any work
- PC10.** lift objects carefully maintaining the correct body posture to avoid injury
- PC11.** use self-rescue apparatus appropriately when required
- PC12.** work in the assigned work zone in the designated work area
- PC13.** follow the recommended safety measures while working at haul roads, heights, dumps, sump areas, stockyards, near moving parts, etc
- PC14.** wear Thermo luminescent Dosimeter (TLD) Badge for personal monitoring in rare earth extraction plant
- PC15.** use caution signs as applicable and obey safety signs, stickers, and tags

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PC16. identify the environmental impact of operations related to mining, mineral separation plant and rare earth extraction plant and take steps to reduce the impact

PC17. check for contamination of sensitive body parts, e.g. face, eyes etc., and seek immediate medical help

Assist in hazard management

To be competent, the user/individual on the job must be able to:

PC18. label and store chemicals appropriately

PC19. display the MSDS for the hazardous chemicals in storage and handling areas

PC20. educate the personnel at the workplace about the safety requirements and follow the applicable safety guidelines

PC21. check the body and clothing for contamination at the end of radioactive work, and decontaminate and take a shower at the decontamination centre

PC22. maintain regular and open communication with co-workers and the employer

Work on equipment and machinery

To be competent, the user/individual on the job must be able to:

PC23. check and ensure every instrument, apparatus and equipment is AERB approved

PC24. operate the appropriate tools, equipment and machinery with proper training and authorization

Check the first aid box, firefighting, and safety equipment

To be competent, the user/individual on the job must be able to:

PC25. check the availability of first aid kits and fire extinguishers, and ensure they are easily accessible

PC26. ensure the availability of emergency showers and eye washers in the chemical handling, unloading and pumping areas

PC27. monitor the work environment at appropriate intervals for hydrogen chloride and hydrocarbons using portable monitors

PC28. operate the appropriate type of fire extinguishers to control different types of fire at the worksite

Report incidents

To be competent, the user/individual on the job must be able to:

PC29. identify and report potential hazards/ unsafe conditions and practices

PC30. follow the appropriate emergency procedure during emergencies, such as power failures, cyclones, etc.

PC31. report the emergencies and any safety incidents as per the organizational procedure

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. different types of chemical plants and their operations

KU2. the importance of time-keeping, discipline and punctuality

KU3. various chemical processes carried out in a rare earth extraction plant

KU4. the impact of violating applicable safety procedures

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- KU5.** the relevant quality norms
- KU6.** different types of individual rare earth chemical processes and the use of associated equipment
- KU7.** different types of effluent treatment processes and the use of associated equipment
- KU8.** the use of different water purifying and waste treatment agents
- KU9.** the hazards and the appropriate safety measures in acid/ alkali handling activities and usage of relevant PPE
- KU10.** hazards and safety aspects involved in acid/ alkali handling activities and usage of relevant PPE
- KU11.** the internal safety organization and the role of the safety committee, workforce inspector and AERB, etc
- KU12.** the importance of sensitization towards different genders and persons with disabilities (PWD)
- KU13.** the Importance of following infection control policies, '5-S' practices, and waste management as prescribed by regulatory authorities like the Atomic Energy Regulatory Board (AERB), United Nations Development Group (UNDG), etc.
- KU14.** the importance of water/material/energy conservation and management
- KU15.** the importance of Social Management Plan (SMP) and Environment Management Plan (EMP) prepared by the organization, as per DGMS guideline
- KU16.** the importance of wearing own TLD or pocket dosimeter (to be obtained from the Health Physics Unit) in the active area
- KU17.** the importance of staying alert and attentive to the work surroundings
- KU18.** the importance of staying in the assigned work zone in the designated work area
- KU19.** the benefits of making the safety rules part of new employee orientation
- KU20.** the importance of contacting the supervisor or manager for instruction, guidance, or training when required

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete appropriate documentation concerning work
- GS2.** make appropriate decisions concerning the area of work with effective reasoning
- GS3.** read and explain manuals, health, and safety instructions, memos, etc
- GS4.** plan and organize work effectively
- GS5.** detect problems at work and use reasoning skills to resolve them promptly and effectively
- GS6.** complete the assigned tasks timely

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain personal health and safety</i>	16	24	-	11
PC1. follow the appropriate hand hygiene measures, i.e., washing hands with EDTA/Soda Ash (Ethylene Diamine Tetra Acetate/Sodium Carbonate), soap and water or alcohol-based hand sanitizers	1	2	-	1
PC2. ensure the plants are provided with a dilution ventilation system to avoid the undue accumulation of chemical vapours in working environments	1	1	-	1
PC3. clean and disinfect the regularly used objects, such as mobile phones, wireless communication systems keys, wallets and work passes	1	1	-	-
PC4. follow the Safety Management Plan (SMP) and Emergency Management Plan (EMP)	1	2	-	-
PC5. follow sanitation measures in the work area, e.g., avoiding eating, drinking, chewing and smoking	1	2	-	-
PC6. use the appropriate PPE, e.g. using an appropriate mask to cover the mouth and nose	1	2	-	-
PC7. identify six directional hazards at the workplace and take decisions accordingly	1	2	-	1
PC8. follow appropriate safety measures to avoid exposure to radiation, e.g. limiting stay in the radiation area and avoiding working with open wounds or exposed body parts in radiation areas	1	2	-	1
PC9. undertake the Take-5 (Personal Risk Assessment) before the commencement of any work	1	2	-	1
PC10. lift objects carefully maintaining the correct body posture to avoid injury	-	1	-	-
PC11. use self-rescue apparatus appropriately when required	1	1	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. work in the assigned work zone in the designated work area	1	1	-	1
PC13. follow the recommended safety measures while working at haul roads, heights, dumps, sump areas, stockyards, near moving parts, etc	1	1	-	1
PC14. wear Thermo luminescent Dosimeter (TLD) Badge for personal monitoring in rare earth extraction plant	1	1	-	1
PC15. use caution signs as applicable and obey safety signs, stickers, and tags	1	1	-	-
PC16. identify the environmental impact of operations related to mining, mineral separation plant and rare earth extraction plant and take steps to reduce the impact	1	1	-	1
PC17. check for contamination of sensitive body parts, e.g. face, eyes etc., and seek immediate medical help	1	1	-	1
<i>Assist in hazard management</i>	5	8	-	4
PC18. label and store chemicals appropriately	1	1	-	1
PC19. display the MSDS for the hazardous chemicals in storage and handling areas	1	2	-	1
PC20. educate the personnel at the workplace about the safety requirements and follow the applicable safety guidelines	1	1	-	-
PC21. check the body and clothing for contamination at the end of radioactive work, and decontaminate and take a shower at the decontamination centre	1	2	-	1
PC22. maintain regular and open communication with co-workers and the employer	1	2	-	1
<i>Work on equipment and machinery</i>	2	4	-	1
PC23. check and ensure every instrument, apparatus and equipment is AERB approved	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. operate the appropriate tools, equipment and machinery with proper training and authorization	1	2	-	1
<i>Check the first aid box, firefighting, and safety equipment</i>	4	10	-	2
PC25. check the availability of first aid kits and fire extinguishers, and ensure they are easily accessible	1	3	-	-
PC26. ensure the availability of emergency showers and eye washers in the chemical handling, unloading and pumping areas	1	2	-	-
PC27. monitor the work environment at appropriate intervals for hydrogen chloride and hydrocarbons using portable monitors	1	2	-	1
PC28. operate the appropriate type of fire extinguishers to control different types of fire at the worksite	1	3	-	1
<i>Report incidents</i>	3	4	-	2
PC29. identify and report potential hazards/ unsafe conditions and practices	1	2	-	-
PC30. follow the appropriate emergency procedure during emergencies, such as power failures, cyclones, etc.	1	1	-	1
PC31. report the emergencies and any safety incidents as per the organizational procedure	1	1	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1810
NOS Name	Follow the applicable health, safety and environmental guidelines (Placer Mining)
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Placer Mining
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

MIN/N1811: Work collaboratively and effectively in a team

Description

This unit is about the importance of teamwork in the workplace.

Scope

The scope covers the following :

- Work effectively with co-workers
- Communicate effectively with co-workers
- Practice inclusiveness at work

Elements and Performance Criteria

Work effectively with co-workers

To be competent, the user/individual on the job must be able to:

- PC1.** identify own direct and indirect responsibilities, e.g. duties concerning team members
- PC2.** undertake appropriate activities to promote cooperation and good relationships with co-workers
- PC3.** mentor and assist subordinates in the execution of their work responsibilities
- PC4.** identify possible disruptions to work through coordination with the relevant stakeholders and take appropriate preventive measures
- PC5.** follow the recommended practices to avoid and resolve conflicts at work
- PC6.** interact effectively and cooperate with co-workers
- PC7.** manage workplace conflicts, ensuring no impact on work
- PC8.** identify and take responsibility for own mistakes
- PC9.** follow the organizational policies and procedures
- PC10.** follow the appropriate productivity enhancement measures, e.g. Quality circle, 5s, TPM, Six Sigma, JIT, etc.
- PC11.** exchange feedback with peers and other team members to improve the quality of work

Communicate effectively with co-workers

To be competent, the user/individual on the job must be able to:

- PC12.** follow the organizational policy for the efficient and timely dissemination of information to the authorized personnel
- PC13.** communicate clearly and politely to ensure effective communication with co-workers
- PC14.** practice active listening to understand the conveyed information and instructions
- PC15.** practice openness in work conversations, giving undivided attention to enhance the quality of workplace communication
- PC16.** maintain the professional standard of communication, e.g. avoiding the use of offensive language at work

Practice inclusiveness at work

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC17.** empathize with Persons with Disabilities (PwD)
- PC18.** practice gender equality at work
- PC19.** maintain respectful behaviour and etiquette at work
- PC20.** listen to and acknowledge the feelings, concerns, opinions, and ideas of others

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the importance of teamwork and the consequence of poor teamwork on project outcomes, timelines, safety, etc
- KU2.** the duties and rights of workers
- KU3.** the importance of creating a healthy and cooperative work environment
- KU4.** the importance of proper and effective communication and the adverse effects of poor communication on quality, timeliness, safety, etc
- KU5.** different modes of communication used at work
- KU6.** the importance and need of supporting co-workers facing problems for the smooth functioning of work
- KU7.** the fundamental concept of gender equality
- KU8.** how to recognize and be sensitive to issues of disability, culture and gender
- KU9.** how to show empathy to PwD appropriately
- KU10.** legislation, policies, and procedures relating to gender sensitivity and cultural diversity including their impact on the area of operation
- KU11.** the appropriate behaviour to be practised with coworkers
- KU12.** the appropriate communication skills and leadership qualities
- KU13.** how to be open to listening to and understanding the other person's point of view

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete appropriate documentation concerning work
- GS2.** make appropriate decisions concerning the area of work with effective reasoning
- GS3.** read and explain manuals, health, and safety instructions, memos, etc
- GS4.** plan and organize work effectively
- GS5.** detect problems at work and use reasoning skills to resolve them promptly and effectively
- GS6.** complete the assigned tasks timely

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Work effectively with co-workers</i>	15	30	-	10
PC1. identify own direct and indirect responsibilities, e.g. duties concerning team members	3	5	-	3
PC2. undertake appropriate activities to promote cooperation and good relationships with co-workers	1	2	-	1
PC3. mentor and assist subordinates in the execution of their work responsibilities	1	2	-	1
PC4. identify possible disruptions to work through coordination with the relevant stakeholders and take appropriate preventive measures	3	4	-	2
PC5. follow the recommended practices to avoid and resolve conflicts at work	1	2	-	1
PC6. interact effectively and cooperate with co-workers	1	1	-	-
PC7. manage workplace conflicts, ensuring no impact on work	1	1	-	-
PC8. identify and take responsibility for own mistakes	1	3	-	1
PC9. follow the organizational policies and procedures	1	3	-	-
PC10. follow the appropriate productivity enhancement measures, e.g. Quality circle, 5s, TPM, Six Sigma, JIT, etc.	1	4	-	1
PC11. exchange feedback with peers and other team members to improve the quality of work	1	3	-	-
<i>Communicate effectively with co-workers</i>	9	10	-	5
PC12. follow the organizational policy for the efficient and timely dissemination of information to the authorized personnel	2	3	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. communicate clearly and politely to ensure effective communication with co-workers	1	2	-	-
PC14. practice active listening to understand the conveyed information and instructions	2	1	-	1
PC15. practice openness in work conversations, giving undivided attention to enhance the quality of workplace communication	2	2	-	1
PC16. maintain the professional standard of communication, e.g. avoiding the use of offensive language at work	2	2	-	1
<i>Practice inclusiveness at work</i>	6	10	-	5
PC17. empathize with Persons with Disabilities (PwD)	2	2	-	2
PC18. practice gender equality at work	2	2	-	2
PC19. maintain respectful behaviour and etiquette at work	1	3	-	1
PC20. listen to and acknowledge the feelings, concerns, opinions, and ideas of others	1	3	-	-
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1811
NOS Name	Work collaboratively and effectively in a team
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Placer Mining
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N1817.Use different material handling equipment in rare earth extraction plant	30	50	0	20	100	20
MIN/N1818.Undertake different unit operations in rare earth extraction plant	30	50	0	20	100	20
MIN/N1819.Undertake the treatment of rare earth extraction plant effluents	30	50	0	20	100	15
MIN/N1820.Follow the Material Safety Data Sheet (MSDS) for chemicals	30	50	0	20	100	15
MIN/N1821.Carry out corrective and preventive maintenance of RE extraction plant equipment	30	50	0	20	100	15
MIN/N1810.Follow the applicable health, safety and environmental guidelines (Placer Mining)	30	50	0	20	100	5
MIN/N1811.Work collaboratively and effectively in a team	30	50	0	20	100	5
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	5
Total	230	380	-	140	750	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.